

GENERAL SCIENCE: BIOLOGY AND PHYSIOLOGY

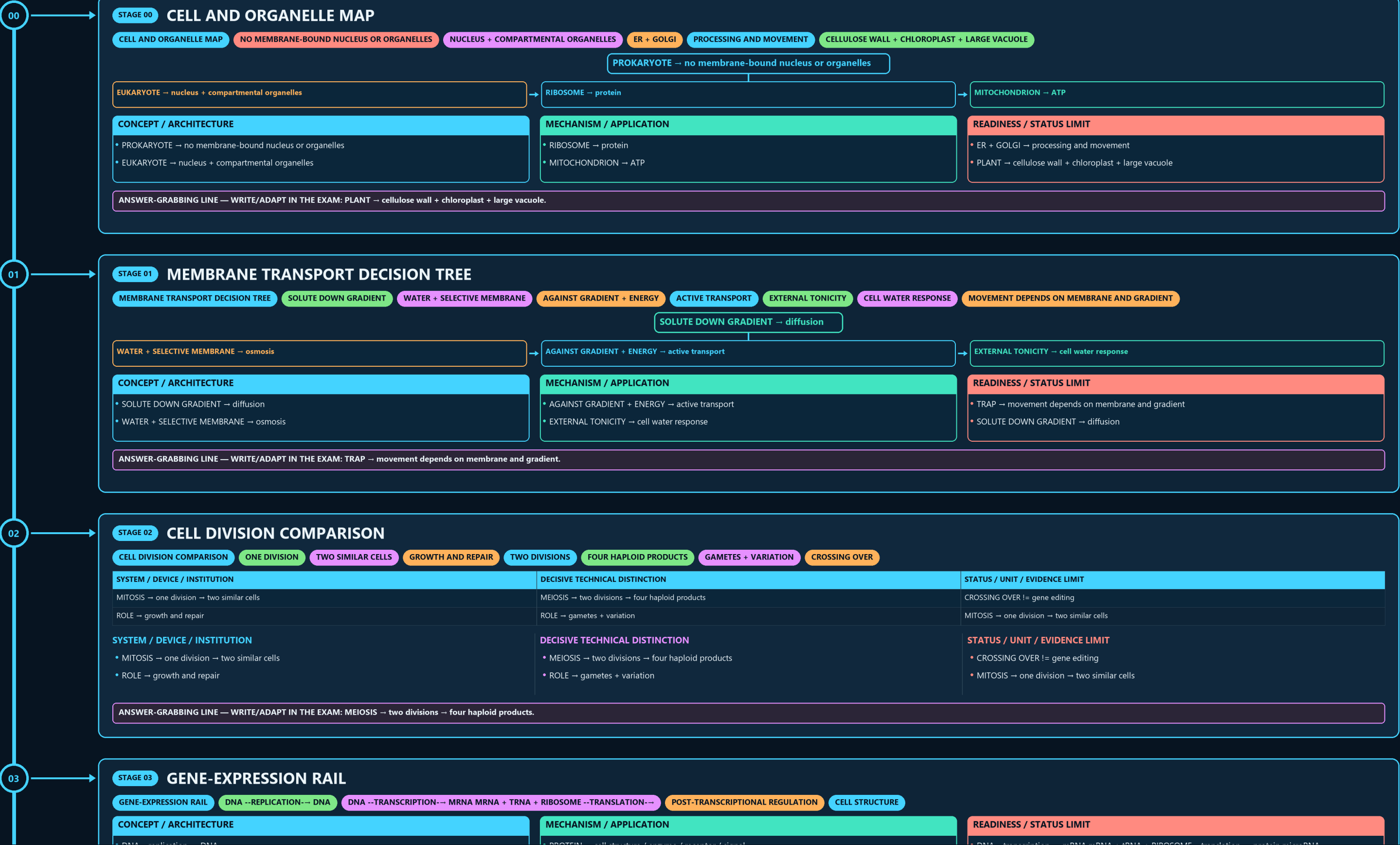
CELL AND ORGANELLE MAP → MEMBRANE TRANSPORT DECISION TREE → CELL DIVISION COMPARISON → GENE-EXPRESSION RAIL → NUTRITION AND DIGESTION ROUTE → TRANSPORT RESPIRATION EXCRETION LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

SYSTEM / DEVICE / INSTITUTION

MITOSIS → one division → two similar cells

ROLE → growth and repair

DECISIVE TECHNICAL DISTINCTION

MEIOSIS → two divisions → four haploid products

ROLE → gametes + variation

STATUS / UNIT / EVIDENCE LIMIT

CROSSING OVER != gene editing

MITOSIS → one division → two similar cells

SYSTEM / DEVICE / INSTITUTION

• MITOSIS → one division → two similar cells

• ROLE → growth and repair

DECISIVE TECHNICAL DISTINCTION

• MEIOSIS → two divisions → four haploid products

• ROLE → gametes + variation

STATUS / UNIT / EVIDENCE LIMIT

• CROSSING OVER != gene editing

• MITOSIS → one division → two similar cells

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEIOSIS → two divisions → four haploid products.

04

STAGE 03

GENE-EXPRESSION RAIL

GENE-EXPRESSION RAIL

DNA --REPLICATION--> DNA

DNA --TRANSCRIPTION--> MRNA MRNA + TRNA + RIBOSOME --TRANSLATION-->

POST-TRANSCRIPTIONAL REGULATION

CELL STRUCTURE

CONCEPT / ARCHITECTURE

• DNA --replication--> DNA

• DNA --transcription--> mRNA mRNA + tRNA + RIBOSOME --translation--> protein microRNA → post-transcriptional regulation

MECHANISM / APPLICATION

• PROTEIN → cell structure / enzyme / receptor / signal

• DNA --replication--> DNA

READINESS / STATUS LIMIT

• DNA --transcription--> mRNA mRNA + tRNA + RIBOSOME --translation--> protein microRNA → post-transcriptional regulation

• PROTEIN → cell structure / enzyme / receptor / signal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROTEIN → cell structure / enzyme / receptor / signal.

05

STAGE 04

NUTRITION AND DIGESTION ROUTE

NUTRITION AND DIGESTION ROUTE

ACID + PEPSIN

FAT EMULSIFICATION

PANCREAS + INTESTINE

DIGESTION COMPLETION

SMALL INTESTINE

PRINCIPAL ABSORPTION

CONCEPT / ARCHITECTURE

• MOUTH → amylase

• STOMACH → acid + pepsin

MECHANISM / APPLICATION

• LIVER / BILE → fat emulsification

• PANCREAS + INTESTINE → digestion completion

READINESS / STATUS LIMIT

• SMALL INTESTINE → principal absorption

• MOUTH → amylase

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PANCREAS + INTESTINE → digestion completion; SMALL INTESTINE → principal absorption.

06

STAGE 05

TRANSPORT RESPIRATION EXCRETION LOOP

TRANSPORT RESPIRATION EXCRETION LOOP

ALVEOLAR GAS EXCHANGE

GASES + NUTRIENTS + HORMONES + WASTES

PULMONARY + SYSTEMIC CIRCULATION

ATP-PRODUCING RESPIRATION

FILTRATION + REABSORPTION + SECRETION

LUNGS → alveolar gas exchange

BLOOD → gases + nutrients + hormones + wastes

HEART → pulmonary + systemic circulation

CELLS → ATP-producing respiration

NEPHRON → filtration + reabsorption + secretion

CONCEPT / ARCHITECTURE

• LUNGS → alveolar gas exchange

• BLOOD → gases + nutrients + hormones + wastes

MECHANISM / APPLICATION

• HEART → pulmonary + systemic circulation

• CELLS → ATP-producing respiration

READINESS / STATUS LIMIT

• NEPHRON → filtration + reabsorption + secretion

• LUNGS → alveolar gas exchange

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOOD → gases + nutrients + hormones + wastes.

STAGE 06

CONTROL AND FEEDBACK LADDER

CONTROL AND FEEDBACK LADDER

CHANGE DETECTED

RAPID WIRED RESPONSE

BLOOD-BORNE HORMONE RESPONSE

PHYSIOLOGICAL ADJUSTMENT

NEGATIVE FEEDBACK

RESTORES REGULATED RANGE

RECEPTOR → change detected

NERVOUS → rapid wired response

ENDOCRINE → blood-borne hormone response

TARGET → physiological adjustment

STARTING CONDITION / INPUT

• RECEPTOR → change detected

• NERVOUS → rapid wired response

SCIENTIFIC OR ENGINEERING MECHANISM

• ENDOCRINE → blood-borne hormone response

• TARGET → physiological adjustment

OUTPUT / FAILURE BOUNDARY

• NEGATIVE FEEDBACK → restores regulated range

• RECEPTOR → change detected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TARGET → physiological adjustment; NEGATIVE FEEDBACK → restores regulated range.

06

STAGE 06 CONTROL AND FEEDBACK LADDER

CONTROL AND FEEDBACK LADDER CHANGE DETECTED RAPID WIRED RESPONSE BLOOD-BORNE HORMONE RESPONSE PHYSIOLOGICAL ADJUSTMENT NEGATIVE FEEDBACK RESTORES REGULATED RANGE

RECEPTOR → change detected

NERVOUS → rapid wired response

ENDOCRINE → blood-borne hormone response

TARGET → physiological adjustment

STARTING CONDITION / INPUT

- RECEPTOR → change detected
- NERVOUS → rapid wired response

SCIENTIFIC OR ENGINEERING MECHANISM

- ENDOCRINE → blood-borne hormone response
- TARGET → physiological adjustment

OUTPUT / FAILURE BOUNDARY

- NEGATIVE FEEDBACK → restores regulated range
- RECEPTOR → change detected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TARGET → physiological adjustment; NEGATIVE FEEDBACK → restores regulated range.

07

STAGE 07 IMMUNITY ARCHITECTURE

IMMUNITY ARCHITECTURE BARRIERS + PHAGOCYTES + COMPLEMENT B CELL T CELL REGULATORY ROLES REGULATORY T CELL + FOXP3 PERIPHERAL TOLERANCE SELF-REACTIVITY CONTROL

CONCEPT / ARCHITECTURE

- BARRIERS + PHAGOCYTES + COMPLEMENT → innate
- B CELL → antibody / memory

MECHANISM / APPLICATION

- T CELL → helper / cytotoxic / regulatory roles
- REGULATORY T CELL + Foxp3 → peripheral tolerance

READINESS / STATUS LIMIT

- SELF-REACTIVITY CONTROL != no immune response
- BARRIERS + PHAGOCYTES + COMPLEMENT → innate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY T CELL + Foxp3 → peripheral tolerance.

08

STAGE 08 DISEASE TRANSMISSION GRID

DISEASE TRANSMISSION GRID CARRIES PATHOGEN, IS NOT DISEASE ANIMAL-HUMAN INTERFACE MECHANISM-SPECIFIC LAYERS

CONCEPT / ARCHITECTURE

VECTOR → carries pathogen, is not disease
ZOOONOSIS → animal-human interface

MECHANISM / APPLICATION

PREVENTION → mechanism-specific layers
VECTOR → carries pathogen, is not disease

READINESS / STATUS LIMIT

ZOOONOSIS → animal-human interface
PREVENTION → mechanism-specific layers

CONCEPT / ARCHITECTURE

- VECTOR → carries pathogen, is not disease
- ZOOONOSIS → animal-human interface

MECHANISM / APPLICATION

- PREVENTION → mechanism-specific layers
- VECTOR → carries pathogen, is not disease

READINESS / STATUS LIMIT

- ZOOONOSIS → animal-human interface
- PREVENTION → mechanism-specific layers

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AIR | WATER/FOOD | BLOOD | CONTACT | VECTOR.

09

STAGE 09 MICROBIAL ECOLOGY AND RESISTANCE

MICROBIAL ECOLOGY AND RESISTANCE ANTIMICROBIAL PRESSURE RESISTANT MICROBE SURVIVAL + SPREAD MATRIX-PROTECTED COMMUNITY CONTEXT-DEPENDENT ROLES EXTREMOPHILE LABEL TOLERANCE, NOT UNIVERSAL SAFETY

ANTIMICROBIAL PRESSURE → selection

RESISTANT MICROBE → survival + spread

BIOFILM → matrix-protected community

MICROBIOME / PROBIOTIC → context-dependent roles

CONCEPT / ARCHITECTURE

- ANTIMICROBIAL PRESSURE → selection
- RESISTANT MICROBE → survival + spread

MECHANISM / APPLICATION

- BIOFILM → matrix-protected community
- MICROBIOME / PROBIOTIC → context-dependent roles

READINESS / STATUS LIMIT

- EXTREMOPHILE LABEL → tolerance, not universal safety
- ANTIMICROBIAL PRESSURE → selection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MICROBIOME / PROBIOTIC → context-dependent roles; EXTREMOPHILE LABEL → tolerance, not universal safety.

10

STAGE 10 PLANT FUNCTION FLOW

